Appendix B: Technical Notes

All stochastic steps use fixed seeds (seed = 42 for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

Data Model I

Each record is a `Paper` dataclass with: title, abstract, authors (list of `Author`), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical\_id} = \begin{cases} \texttt{doi:} + \text{normalize}(\text{DOI}) & \text{if DOI present} \\ \texttt{arxiv:} + \text{arXiv\_id} & \text{if arXiv ID present} \\ \texttt{s2:} + \text{S2\_id} & \text{if S2 ID present} \\ \texttt{openalex:} + \text{OpenAlex\_id} & \text{if OpenAlex ID present} \\ \texttt{title:} + \text{SHA1}(\text{title})[:16] & \text{otherwise} \end{cases}$$

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

Non-negative matrix factorization decomposes the TF-IDF document-term matrix $\mathbf{V} \in \mathbb{R}^{m \times n}$ (where m is the number of documents and n is the vocabulary size) into $\mathbf{W} \in \mathbb{R}^{m \times k}$ and $\mathbf{H} \in \mathbb{R}^{k \times n}$, where k is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W},\mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{W}\mathbf{H}\|_F^2$$

using multiplicative update rules (Lee and Seung 1999) with a fixed random seed for reproducibility. The topic-term matrix $\mathbf{H}$ gives the top terms per topic; the document-topic matrix $\mathbf{W}$ gives each document's topic loadings.

Growth Rate Estimation I

The compound annual growth rate is:

$$\mathsf{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}}\right)^{1/(T_{\text{end}}-T_{\text{start}})} - 1$$

where N_{start} and N_{end} are the publication counts in the first and last years of the corpus, respectively. The doubling time is $t_d = \ln(2)/\ln(1+\mathsf{CAGR})$. For this run: CAGR = 6.76%, doubling time = 2.1 years.

Configuration Surface I

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 4 engines, a 4-bucket taxonomy, and 4 hypotheses.

Intermediate and final outputs live under `output/` and are disposable and regenerable:

Artifacts I

File	Stage	Description
`corpus.jsonl`	01	De-duplicated corpus (46 records)
`temporal_analysis.json`	02	Year counts, CAGR, doubling time, peak year
`subfield_classification.json`	02	Per-bucket paper counts
`subfield_timeline.json`	02	Per-subfield annual breakdown
`tfidf_data.json`	02	TF-IDF matrix, feature names, doc tokens
`topics.json`	02	NMF topic-term distributions
`citation_network.json`	02	Network metrics, PageRank, HITS, communities
`citation_graph.gml`	02	GraphML citation graph
`nanopublications.jsonl`	03	LLM-extracted assertions (0 in this run)
`hypothesis_scores.json`	03	Per-hypothesis evidence scores
`cross_phase_analysis.json`	01 / 03	Phase membership, overlap, citation sufficiency, and explicit hypothesis-score claim boundary
`fulltext_assessment.json`	06	Abstract/OA/PDF coverage report